

Theory of Matting

CS5245 Vision & Graphics for Special Effects

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Theory of Matting

Basic problem of blue screen matting:

Given an image of a foreground object shot in front of a backing color (blue screen or green screen or other colors), obtain a matte of the foreground object so that the foreground object can be blended into a new background image using the matte to produce a new composite image.

Smith and Blinn presented a very good foundation for blue screen matting techniques [SB96].

Let $C = [R, G, B]$ denote a color with $0 \leq R, G, B \leq 1$.
Let α denotes a transparency value with $0 \leq \alpha \leq 1$.

Notations:

- Foreground image color: $C_f = [R_f, G_f, B_f], \alpha_f = 1$
- Backing (screen) color: $C_k = [0, 0, B_k], \alpha_k = 1$
(assuming blue screen)
- Original foreground object color: $C_o = [R_o, G_o, B_o]$
- Background image color: $C_b = [R_b, G_b, B_b], \alpha_b = 1$
- Composite image color: $C_c = [R_c, G_c, B_c]$

Then, the matting problem can be defined formally as follows:

Given C_f and C_b at corresponding pixels, and C_k a known backing color, and assuming $C_f = \alpha_o C_o + (1 - \alpha_o) C_k$, determine α_o and C_o , which then gives the composite color $C_c = \alpha_o C_o + (1 - \alpha_o) C_b$ at the corresponding point, for all points that C_f and C_b share in common.

Note:

- The notations used here are simpler and are slightly different from those in [SB96].

From the problem definition, we have the following equations:

$$R_f = \alpha_o R_o \quad (1)$$

$$G_f = \alpha_o G_o \quad (2)$$

$$B_f = \alpha_o B_o + (1 - \alpha_o) B_k. \quad (3)$$

So, we have 4 unknowns (R_o , G_o , B_o , α_o) but only 3 equations!
Cannot get a unique solution in general!

But, why is it the blue screen technique has been successfully applied in the movie industry?

Reason: A unique solution exists under special conditions.

Case 1: No Blue

- There is no blue in C_o , i.e., $B_o = 0$.
- And $B_k \neq 0$.

In this case, there are only 3 unknowns, R_o , G_o , α_o , with 3 equations.

Then, from Eq. 3, we obtain

$$\alpha_o = 1 - \frac{B_f}{B_k}. \quad (4)$$

Note: $B_k \neq 0$ and $B_f \leq B_k$.

Then,

$$R_o = \frac{1}{\alpha_o} R_f, \quad G_o = \frac{1}{\alpha_o} G_f. \quad (5)$$

Notes:

- This case is very restrictive.
- It rules out many colors, including grays because grays have blue.

Case 2: Gray and Skin Color

- $R_o = aB_o$ for some known a (or $G_o = aB_o$).
- And $B_k \neq 0$.

Then, from Eqs. 1 and 3, we obtain

$$\alpha_o = \frac{R_f - aB_\Delta}{aB_k} \quad (6)$$

where $B_\Delta = B_f - B_k$.

Then,

$$R_o = \frac{1}{\alpha_o} R_f, \quad G_o = \frac{1}{\alpha_o} G_f. \quad (7)$$

Example 1: Gray

$R_o = G_o = B_o$, i.e., $a = 1$.

Then,

$$\alpha_o = \frac{R_f - B_\Delta}{B_k} \quad (8)$$

This case applies to science fiction movie in which the spaceships are mostly gray.

Example 2: Skin Color

$R_o \approx 2G_o \approx 2B_o$ for all races.

Then,

$$\alpha_o = \frac{R_f - 2B_\Delta}{2B_k} \quad (9)$$

This case applies to human faces, hands, legs, etc.

General Matting Problems

To solve the matting problem in general:

- Need to take the same image with two different backing colors.
- This gives four equations for solving the four unknowns.

Case 1: Use Two Different Shades of Blue

Use two different shades of blue B_{k1} and B_{k2} as backing colors.

$$\begin{aligned}C_{f1} &= \alpha_o C_o + (1 - \alpha_o) B_{k1} \\C_{f2} &= \alpha_o C_o + (1 - \alpha_o) B_{k2} .\end{aligned}\tag{10}$$

Then,

$$\begin{aligned}B_{f1} &= \alpha_o B_o + (1 - \alpha_o) B_{k1} \\B_{f2} &= \alpha_o B_o + (1 - \alpha_o) B_{k2} .\end{aligned}\tag{11}$$

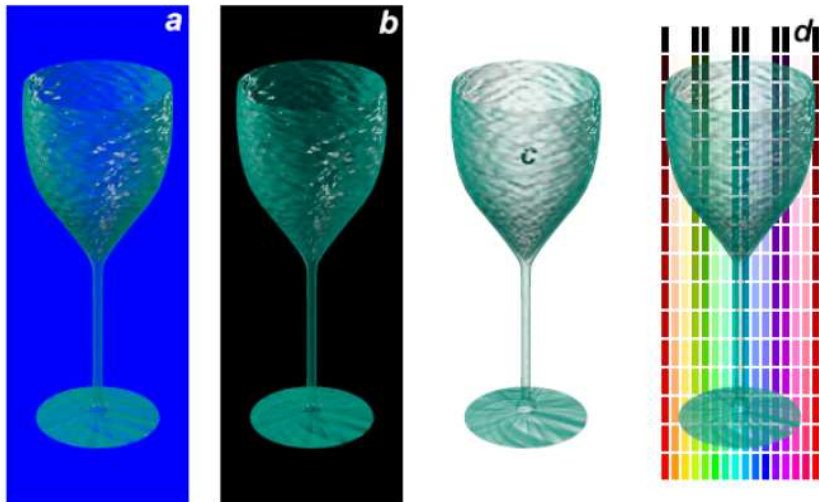
Subtracting the two equations yields

$$B_{f1} - B_{f2} = (1 - \alpha_o)(B_{k1} - B_{k2}) .\tag{12}$$

Then,

$$\alpha_o = 1 - \frac{B_{f1} - B_{f2}}{B_{k1} - B_{k2}} .\tag{13}$$

Now, can solve for R_o, G_o, B_o .



- (a, b) Images taken with two different backing colors.
- (c) Pulled foreground.
- (d) Composited against new background.

Case 2: Use Two Different Backing Colors

Use two different backing colors C_{k1} and C_{k2} .

$$\begin{aligned}C_{f1} &= \alpha_o C_o + (1 - \alpha_o) B_{k1} \\C_{f2} &= \alpha_o C_o + (1 - \alpha_o) B_{k2} .\end{aligned}\tag{14}$$

Subtracting the two equations yields

$$C_{f1} - C_{f2} = (1 - \alpha_o)(C_{k1} - C_{k2}) .\tag{15}$$

This set of equation is over-constrained: 1 unknown, 3 equations.

Solution 1: Add up the three equations.

Then,

$$\alpha_o = 1 - \frac{(R_{f1} - R_{f2}) + (G_{f1} - G_{f2}) + (B_{f1} - B_{f2})}{(R_{k1} - R_{k2}) + (G_{k1} - G_{k2}) + (B_{k1} - B_{k2})} \quad (16)$$

provided the denominator is not 0.

Solution 2: Apply Least Squares Method

Let's simplify notation. Define:

- $C_{\Delta} = C_1 - C_2$
- $\beta = 1 - \alpha_o$

Then, Eq. 15 can be written as

$$\begin{aligned}\beta R_{k\Delta} - R_{f\Delta} &= 0 \\ \beta G_{k\Delta} - G_{f\Delta} &= 0 \\ \beta B_{k\Delta} - B_{f\Delta} &= 0.\end{aligned}\tag{17}$$

Sum of squared error E of the above equations is

$$E = (\beta R_{k\Delta} - R_{f\Delta})^2 + (\beta G_{k\Delta} - G_{f\Delta})^2 + (\beta B_{k\Delta} - B_{f\Delta})^2.\tag{18}$$

To obtain the least squares solution, set $dE/d\beta = 0$ and solve for the optimum β .

Then, we obtain

$$\alpha_o = 1 - \frac{R_{f\Delta}R_{k\Delta} + G_{f\Delta}G_{k\Delta} + B_{f\Delta}B_{k\Delta}}{R_{k\Delta}^2 + G_{k\Delta}^2 + B_{k\Delta}^2}. \quad (19)$$



Can differentiate between transparent and opaque objects that both appear to be blue.

Matting Without Blue Screen

A method proposed by Ruzon and Tomasi [RT00].

- User specify object region and boundary region.
- Alpha value of object region is set to 1.
- Alpha value of boundary region is computed by estimating the contributions of neighboring objects' colors.



Chuang et al. [CCSS01] apply a similar idea but use Bayesian approach to estimate alpha.

Input



Segmentation

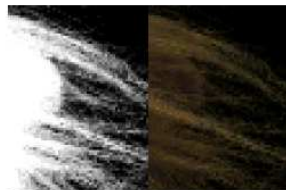


Composite



Comparison with Ruzon and Tomasi's method:

Ruzon and Tomasi



Bayesian approach



Ground truth



Alpha Matte

Composite

Inset

Shadow Matting

Chuang et al. [CGC⁺03]:

- Pull a matte of shadow.
- Acquire photometric and geometric properties of the target scene by sweeping oriented linear shadows across it.
- Then, composite the shadow onto the scene.



(a) Source frame



(b) Target frame



(c) Composite

Environmental Matting

Zongker et al. [ZWCS99] introduced environment matting problem:

- Use structured light patterns to capture transparency as well as refraction and reflection properties.



Left: alpha matte. Middle: environment matte. Right: photo.

Reference I

-  Y.-Y. Chuang, B. Curless, D. H. Salesin, and R. Szeliski.
A bayesian approach to digital matting.
In Proc. IEEE CVPR, pages II-264–II-271, 2001.
-  Y.-Y. Chuang, D. B. Goldman, B. Curless, D. H. Salesin, and R. Szeliski.
Shadow matting and compositing.
ACM Transactions on Graphics, 22(3):494–500, July 2003.
-  M. A. Ruzon and C. Tomasi.
Alpha estimation in natural images.
In Proc. IEEE CVPR, pages 18–25, 2000.
-  A. R. Smith and J. F. Blinn.
Blue screen matting.
In Proc. ACM SIGGRAPH, pages 259–268, 1996.

Reference II



D. E. Zongker, D. M. Werner, B. Curless, and D. H. Salesin.
Environment matting and compositing.
In *Proc. SIGGRAPH*, pages 205–214, 1999.